

AYMANE AIT DADS

Data Science Intern | Machine Learning Engineering · Computer Vision · LLM Fine-Tuning

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by designing and executing an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer model under real GPU-constrained conditions. Reduced manual network analysis time by 60% at Orange Maroc by building production-ready Random Forest and K-Means pipelines on 100K+ fiber-optic records, delivering actionable maintenance recommendations to stakeholders. Brings hands-on expertise in PyTorch, computer vision, NLP, and ML pipeline engineering - skills directly aligned with DZYNE's need for data-driven autonomy and defense-grade system development. Eager to apply ablation-driven ML engineering and multimodal modeling to DZYNE's autonomous defense platforms operating across 50+ countries.

CORE COMPETENCIES

Machine Learning Engineering | Computer Vision | Data Pipeline Development | Autonomous Systems | Model Evaluation & Ablation | Predictive Modeling | Deep Learning Frameworks | Feature Engineering

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built end-to-end **machine learning pipeline** combining Random Forest and K-Means on 100K+ fiber-optic network records - upload speed, latency, jitter - to classify quality across thousands of nodes.
- Designed a **clustering pipeline** that mapped customer performance profiles to 5 commercial service tiers, with outputs adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **60%** through automated feature engineering, model evaluation, and reproducible data processing workflows built in Python, Scikit-learn, and Pandas.
- Delivered stakeholder presentations translating **model outputs** into maintenance recommendations, bridging technical findings and operational decision-making for a non-technical audience.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge - Kaggle (In Progress)

Kaggle Competition · 2025-2026 · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) by fine-tuning a 30B-parameter Mamba-Transformer with LoRA SFT ($r=32$, BF16, completion-only loss, 8,192-token context, all attention + MLP + lm_head projections targeted) under strict GPU constraints.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · PEFT · TRL · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection - NTIRE 2026 @ CVPR

EURECOM + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in EURECOM Class

- Ranked **1st in EURECOM class** (0.791 AUC, private Kaggle leaderboard) by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head on 250K images spanning 25+ generator types using FP16 mixed precision and dual learning rates.
- Submitted to **NTIRE 2026 @ CVPR CodaBench** international benchmark; discovered that 1-epoch training (0.793 AUC) outperforms 4-epoch training (0.767 AUC), demonstrating stronger out-of-distribution generalization via early stopping combined with 10-view TTA ensemble.

PyTorch · OpenCLIP · ViT · AdamW · Kaggle Notebooks

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Image Security, Statistics, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Scikit-learn | Hugging Face Transformers | Unsloth | TRL | PEFT | vLLM

ML Methods & Domains: LoRA SFT | Computer Vision | NLP | Ensemble Methods | Feature Engineering | Ablation Studies | Clustering

Data & Infrastructure: Python | SQL | Pandas | NumPy | Power BI | Git | Docker | Linux